Md. Didarul Islam

PhD Student | Doctoral Researcher | Mechanical and Aerospace Engineering

Flow Information Lab, Gyeongsang National University, Jinju-si, South Korea

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Professional Profile

PhD student and doctoral researcher in Mechanical and Aerospace Engineering at Gyeongsang National University, South Korea, with research experience in computational fluid dynamics, biofluid engineering, turbomachinery, composite materials, and machine learning-assisted engineering analysis. Research work integrates CFD, experimental validation, reduced-order modeling, and data-driven methods for biomedical and practical engineering applications, including upper-airway airflow, obstructive sleep apnea diagnosis, ureteral stent flow obstruction, pseudoaneurysm treatment flow, and pump/propeller systems.

Research Interests

- Computational Fluid Dynamics (CFD), biofluid engineering, and medical-twin modeling
- Upper-airway airflow, obstructive sleep apnea diagnosis, and breathing power/diagram features
- Turbomachinery, pump systems, propeller performance, cavitation, erosion, and hydraulic optimization
- Composite materials, natural fiber-reinforced composites, and biomedical/structural materials
- Machine learning, Dynamic Mode Decomposition, Koopman theory, and reduced-order modeling

Education

Doctor of Philosophy (Ph.D.), Department of Mechanical and Aerospace Engineering. - Ongoing *2025-2028*
Gyeongsang National University, South Korea

- Research project: Development of a 3D Upper Airway Medical Twin based on Reduced-order Model (RoM) for non-invasive diagnosis of obstructive sleep apnea.

Master of Science (M.Sc.) in Mechanical Engineering, *2023-2025*
Gyeongsang National University, South Korea

- CGPA: 4.21/4.50. Thesis: Experimental and Computational Fluid Dynamics Analysis of Thrombin-Induced Coagulation in Complex Morphologies of Femoral Artery Pseudoaneurysms. (Link)

Master of Science (M.Sc.) Coursework in Mechanical Engineering, *2019-2021*
Dhaka University of Engineering & Technology, Bangladesh

- CGPA: 3.83/4.00. Thesis proposal: Mechanical and thermal characterization of silicon matrix composites.

Bachelor of Science (B.Sc.) in Mechanical Engineering, *2014-2017*
IUBAT— International University of Business Agriculture and Technology, Bangladesh

- CGPA: 3.84/4.00; ranked 1st out of 92. Practicum project on refrigerator production processes. (Link)

Research Experience

Doctoral Researcher, *Sep 2025-Aug 2028*
Flow Information Lab, Gyeongsang National University, South Korea

- Conducting research on 3D upper-airway medical-twin modeling, reduced-order modeling, CFD analysis, and non-invasive diagnosis of obstructive sleep apnea.
- Working on CFD-to-data pipelines for physiological airflow, diagnostic feature extraction, and validation against clinical or simulation-based indicators.

Graduate Research Assistant, *Sep 2023-Aug 2025*
Flow Information Lab, Gyeongsang National University, South Korea

- Performed experimental and computational analyses of thrombin-induced coagulation and biofluid behavior in femoral artery pseudoaneurysm morphologies.
- Contributed to CFD and experimental research on ureteral stent flow impediments and encrustation using an in vitro urinary tract model.

Undergraduate Thesis Supervisor, *Jan 2021-Aug 2023*
Department of Mechanical Engineering, IUBAT, Bangladesh

- Supervised undergraduate research projects and supported students in experimental planning, engineering analysis, and academic reporting.

Teaching and Industry Experience

Senior Lecturer - Department of Mechanical Engineering, IUBAT, Bangladesh *Jan 2022-Aug 2023*
- On study leave for doctoral studies. Teaching and academic responsibilities in mechanical engineering.

Lecturer - Department of Mechanical Engineering, IUBAT, Bangladesh *Jan 2020-Dec 2021*
- Delivered undergraduate-level teaching, laboratory instruction, and student mentoring in mechanical engineering subjects.

Assistant Engineer (Mechanical) - Shams Engineering, Bangladesh *Jan 2019-Dec 2019*
- Supported mechanical engineering design, analysis, maintenance, and project-related technical reporting.

Instructor - Pubergaon Polytechnic and Medical Institute, Bangladesh *Jan 2018-Dec 2018*
- Delivered technical instruction and practical training for engineering students.

Internship Trainee - Minister Hi-Tech Park, Bangladesh *Sep 2017-Dec 2017*
- Completed industrial training related to refrigerator production processes and manufacturing workflow.

Selected Research Projects

- **Development of a 3D Upper Airway Medical Twin based on Reduced-order Model for non-invasive diagnosis of obstructive sleep apnea**
  - Glocal R&D Research Center (2025.09-2028.08).
- **Global Aerospace Digital Composites Pioneering Research**
  - GADIST Research Center, Gyeongsang National University (2025.03-2027.02).
- **Rim-driven pump-equipped firefighting drone for high-rise building fire response**
  - College of Space and Aeronautics Research Center, GNU (2025.04-2026.02).
- **Small hydropower by-pass piping CFD analysis data**
  - K-water (2025.02-2025.12).
- **Optimal thrombin injection method for femoral artery pseudoaneurysm treatment**
  - National Research Foundation of Korea (2023.09-2025.02).

Selected Publications

1. Experimental and CFD analysis of flow impediments and encrustation in ureteral stents using in vitro urinary tract model.
   Islam, M. D., Kang, H. J., Tae, S. J., Kim, K. W., Choi, Y. H., Lee, S. B .., & Kim, H. H. *Scientific Reports* (2025). DOI
2. A new approach to evaluate obstructive sleep apnea according to body mass index using breathing diagram.
   Islam, M. D., Kim, J. S., Jeon, S. J., Kang, H. J., Kim, K. W., Jeon, M. G., et al. *Physics of Fluids* (2024). DOI
3. Koopman theory-based reduced-order modeling for diagnosing obstructive sleep apnea using breathing power diagram.
   Kang, H.*, **Islam, M. D.***, Choi, Y. H., Kim, H. H., & Koo, B. *Physics of Fluids* (2026). DOI
4. Back-turn Approach for Optimal Operation of Booster Pump Systems.
   Bae, Y. S., **Islam, M. D.**, Choi, Y. D., Kang, H. J., Tae, S. J., & Kim, H. H. *Journal of Applied Fluid Mechanics* (2025). DOI
5. Numerical Simulation of Solid-Liquid Two-Phase Flow Analysis of Submersible Drainage Pumps.
   Rakibuzzaman, M., **Islam, M. D.**, Kim, H. H., Suh, S. H., Zhou, L., & Iqbal, A. P. *Irrigation and Drainage* (2025). DOI
6. Analysis of epoxy composites reinforced with jute, banana, and coconut fibers and enhanced with Rubik's layer.
   Al Mahmud, M. Z., **Islam, M. D.**, & Rabbi, S. F. *Journal of the Mechanical Behavior of Biomedical Materials* (2023). DOI

Selected Conference Presentations

- **CFD Analysis of Duct Shape Effects on Thrust in Electric Hubless Rim-Driven Propellers**.
  Korean Society of Fluid Machinery Winter Conference, South Korea (2025).
- **Back-Turn method for energy optimization in Booster pump systems**.
  5th IAHR Asian Working Group Symposium on Hydraulic Machinery and Systems, South Korea (2025).
- **Effects of Cavitation and Erosion on the Hydraulic Performances of SDPs**.
  4th Asian Workshop on Hydraulic Machinery, Japan (2025).

Technical Skills

CAD / Geometry: AutoCAD, SolidWorks, NX, Ansys SpaceClaim

Simulation / CFD: Ansys CFX, ICEM CFD, Ansys Workbench, CFD/FEM analysis

Programming: Python, C, Google Colab

Numerical / Data Analysis: MATLAB, RStudio, Minitab, MS Excel

Reference Management: Mendeley, EndNote, Zotero

Imaging / Diagrams: NIH BioArt, 3D Doctor, ImageJ, Draw.io, Canva

Awards and Scholarships

- Best Presentation Award - IAHR-Asia 2025 Conference, South Korea (2025)
- Young Pioneer Researcher Award - Gyeongsang National University, South Korea (2025,2026)
- Miyan Publications Rewards - IUBAT, Bangladesh (2024,2025,2026)
- Academic Excellence Award / Dean's List - IUBAT, Bangladesh (2015-2018)
- GNU Fellowship - Gyeongsang National University, South Korea (2025-2028)
- CSA Research Scholarship - Gyeongsang National University, South Korea (2025-2026)
- GNU Fellowship - Master's study, Gyeongsang National University (2023-2025)
- Merit-Based Scholarship, 100% Tuition Fee Waiver - IUBAT, Bangladesh (2014-2017)

Professional Memberships

- Student Member - Korean Society of Fluid Machinery (KSFM)
- Student Member - Korean Society of Mechanical Engineers (KSME)
- Student Member - Biomedical Engineering Society for Circulatory Disorders (BESCO)

References

1. **Prof. Hyoung-Ho Kim**
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   Associate Professor,
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   Gyeongsang National University. South Korea
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2. **Prof. A. K. M. Parvez Iqbal**
   Ph.D., University Putra Malaysia (UPM)
   Professor & Coordinator,
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3. **Prof. Md. Abdul Karim**
   Ph.D., Tennessee Tech University, United States
   Assistant Professor,
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